

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Abstract—Automated Personal Protective Equipment (PPE) compliance monitoring via CCTV is vital for injury prevention on construction sites. However, edge surveillance faces severe operational bottlenecks: distant tiny helmet targets ($< 20 \times 20$ pixels), extreme category skew (1 : 12 helmet-to-person ratio), multi-scale occlusions, and background clutter. We present Rep-YOLO11s, a domain-specialized detector engineered as a principled Hardware-Software Co-Design for real-time edge surveillance. Guided by the GPU Roofline model, our architecture resolves the memory-bandwidth bottlenecks of single-batch inference while targeting four core physical constraints of construction monitoring: (1) Memory Access Cost (MAC) minimization via Structural Re-parameterization (RepConv) multi-branch training blocks folded algebraically into single 3×3 convolutions at inference; (2) vertical spatial inductive priors via Stem CoordConv (+0.06 ms) resolving ground distractor confusion; (3) Bi-Level Routing Attention (BiFormer, $S = 8, k = 4$) pruning 93.75% of irrelevant background tokens in neck stages (+0.24 ms) without quadratic complexity; and (4) decoupled multi-task loss pairing Focal EIoU Loss for boundary quality with Task-Aligned Assignment (TAL) for severe 1:12 class imbalance. On the Safety Helmet Wearing Dataset (SHWD / VOC2028), Rep-YOLO11s achieves 94.83% mAP_{50} and 62.54% mAP_{50-95} under joint two-class evaluation, with 5-fold cross-validation reaching a mean mAP_{50} of 96.64% (standard deviation 0.32%, peak 97.11%). Under the Harmonized Hat-Only Protocol, it achieves 77.90% mAP_{50-95} at 640×640 and 78.93% at 960×960 (+4.80% over baseline). In matched-engine deployment audits, batch-one TensorRT FP16 execution at 640×640 achieves 2.92 ms (342.5 FPS) on NVIDIA Tesla T4, 4.37 ms (228.6 FPS at 28.4 W) on RTX 3050 Laptop GPU (182.4 FPS / 12.16 Frames/J under a 15 W power cap), and real-time execution on an entry-level GeForce MX230 (36.0 ms, 27.8 FPS). Zero-shot evaluations across GDUT-HWD, Hard Hat Workers, and SHD confirm robust cross-domain transferability under harmonized PPE protocols.

Index Terms—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.

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I. INTRODUCTION

Personal Protective Equipment (PPE) compliance monitoring, particularly for safety helmets, is strictly mandated across industrial construction sites to prevent traumatic head injuries. While automated intelligent video analytics via CCTV streams offer continuous, non-intrusive compliance auditing, deploying computer vision models in real-world construction environments encounters four major operational bottlenecks: (1) **Distant Tiny Targets**: long-range CCTV cameras render helmets as sub- 20×20 pixel targets subject to severe feature attenuation; (2) **Severe Category Skew**: industrial benchmarks like SHWD exhibit acute category skew (9,044 helmet vs. 111,514 person instances, a 1 : 12 ratio), biasing standard losses; (3) **Spatial Distractor Confusion**: round yellow buckets and orange cones trigger false alarms under translation-invariant convolutions lacking vertical context; and (4) **Strict Edge Latency Bounds**: multi-camera streams require real-time throughput (≥ 60 FPS), penalizing complex attention with high Memory Access Cost (MAC).

To overcome these barriers, we formulate **Rep-YOLO11s** as a principled Hardware-Software Co-Design for edge video surveillance guided by the GPU Roofline model. Rather than treating architectural enhancements as disconnected heuristics, our design directly targets the memory-bandwidth bottleneck of single-stream ($N_{\text{batch}} = 1$) edge inference, where operational intensity (18–28 FLOPs/byte) sits an order of magnitude below the compute ceiling. Our scientific contributions are: (1) **Roofline-Guided RepConv Structural Fusion**: Multi-branch training topologies (3×3 , 1×1 , and identity strictly guarded by $\mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}}$) fold algebraically into single 3×3 kernels (switch_to_deploy) for inference, eliminating intermediate DRAM roundtrips and multi-kernel launch overheads. (2) **Stem CoordConv Inductive Priors**: Normalized coordinates $(x, y) \in [-1, 1]$ injected strictly at the initial Stem layer ($5 \rightarrow 64$) supply vertical positional priors at negligible cost (+0.06 ms), differentiating head-worn helmets from ground distractors without intermediate tensor inflation. (3) **Dynamic Sparse Attention with Native TRT Export**:

Dynamic bi-level routing attention ($S = 8, k = 4$) in neck stages (P_4, P_5) filters out 93.75% of irrelevant background tokens, exported directly to ONNX Opset 17 via native TopK/Gather without custom C++ plugins (+0.24 ms). (4) **Decoupled Multi-Task Imbalance Supervision:** Bounding box regression quality imbalance is resolved via Focal EIoU ($\gamma = 0.5$), which eliminates aspect-ratio gradient freezing, while acute 1 : 12 category imbalance is decoupled and handled by Task-Aligned Assignment (TAL) and aligned classification loss. (5) **Matched-Engine Empirical Audits:** Validated on SHWD (94.83% mAP_{50} ; 5-fold CV 96.64%) and three external domains under harmonized protocols. Benchmarked with strict stream synchronization across Tesla T4, RTX 3050 Laptop GPU, and entry-level MX230 across matched PyTorch and TensorRT FP16 engines.

II. RELATED WORK

A. Real-Time Object Detection Frameworks

Real-time single-stage object detection has evolved through the YOLO series [1]–[3]. YOLOv6 [4] introduced structural re-parameterization via EfficientRep backbones and Rep-PAN. YOLOv8 adopted anchor-free C2f modules, while YOLOv10 [2] explored dual-assignment NMS-free training. Ultralytics YOLO11 [3] refined pyramid alignment via C3k2 and C2PSA blocks. Recently, YOLO26 [5] explored dual-label assignment for NMS-free and DFL-free edge deployment, while Visual State Space Models (e.g., VMamba [6]) adopt linear sequence modeling for vision. However, in dense construction scaffolding with severe occlusion, NMS-free one-to-one matching risks recall degradation on overlapping tiny helmets, while 1D sequence serialization in SSMs sacrifices 2D spatial locality unless augmented with multi-directional scans. Rep-YOLO11s couples Task-Aligned Assignment (TAL) with Bi-Level Routing Attention (BiFormer) to preserve 2D local context while bounding latency.

B. Safety Helmet Detection in Complex Environments

Various architectures have tailored detectors for PPE auditing. EC-YOLOv8 [7] incorporated ECA attention, CARAFE upsampling, and EIoU loss into YOLOv8n (95.7% mAP_{50} on SHWD), but CARAFE reduces speed to 172 FPS on desktop GPUs and remains sensitive to distractors. Wu *et al.* [8] proposed YOLO-CBF combining CoordConv, BiFormer attention, and Focal-EIoU on a small 962-image road dataset without validating against industrial clutter. Zhao *et al.* [9] used DCNv2 and RCF for power-grid inspection, but DCNv2 incurs irregular memory access that penalizes TensorRT. Song *et al.* [10] combined DWR modules, ASPP, and NWD loss, but parallel dilated branches expand memory footprint. Additional small-object architectures including DABFNet [11], MAF-YOLO [12], and TinyFormer [13] explore attention and multi-scale pyramids. MAF-YOLO achieves 94.20% mAP_{50} on helmets via Rep-HELAN but retains un-fused parallel branches at inference (31.4 GFLOPs, 124 FPS), while TinyFormer incurs quadratic token complexity (78 FPS). In contrast, Rep-YOLO11s confines RepConv to critical neck paths

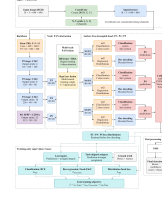


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv coordinate concatenation ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone feature extraction (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck abstraction with Bi-Level Routing Attention (BRA) and RepConv; (4) Anchor-free decoupled detection heads (P3, P4, P5) predicting classification logits ($cv3$) and distribution regression ($cv2$); and (5) Multi-task loss supervision with Task-Aligned Assigner (TAL), BCE, Focal-EIoU, and DFL.

and folds branches into single 3×3 convolutions at deployment (22.4 GFLOPs, 342.5 FPS), eliminating Memory Access Cost (MAC) while pruning 93.75% of background tokens via BiFormer.

C. Structural Re-parameterization and Spatial Encoding

Structural re-parameterization, pioneered by RepVGG [14] and adopted in YOLOv6 [4], decouples training representation from inference execution. Parallel branches (3×3 , 1×1 , identity) provide gradient diversity during backpropagation and fold algebraically into single 3×3 convolutions at deployment. While YOLOv6 applies RepConv indiscriminately across all backbone stages, which exacerbates memory traffic in early high-resolution feature maps, Rep-YOLO11s restricts RepConv strictly to neck aggregation stages and injects CoordConv [15] at the stem, preserving efficiency while providing vertical spatial priors critical for distinguishing helmets on human heads from ground-level construction materials.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with Stem CoordConv; (2) a Path Aggregation Network (PAN) Neck [17], [18] integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve high non-linear feature representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During the training phase (Fig. 2), a RepConv layer with input channels C_{in} , output channels C_{out} , and stride s is formulated across three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}} \cdot \text{BN}_{id}(x), \quad (1)$$

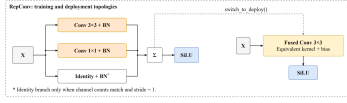


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [14]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

where $W^{3 \times 3} \in \mathbb{R}^{C_{out} \times C_{in} \times 3 \times 3}$ and $W^{1 \times 1} \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$ represent convolution kernels, and BN denotes batch normalization parameters $\{\mu, \sigma^2, \gamma, \beta\}$. Crucially, the identity mapping branch $BN_{id}(x)$ exists if and only if $C_{in} = C_{out}$ and stride $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly ensuring dimensional and stride consistency without requiring additional projection convolutions.

During deployment conversion (switch_to_deploy), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b' :

$$W'_{i,:,:,} = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_{i,:,:,}, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}. \quad (2)$$

Next, $W'_{1 \times 1}$ is zero-padded into a 3×3 kernel $W'_{1 \times 1 \rightarrow 3 \times 3}$ at center position (2, 2). When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into an equivalent 3×3 Dirac delta kernel W'_{id} where $W'_{id,c,c,2,2} = 1$ and 0 elsewhere, folded with BN_{id} ; when $C_{in} \neq C_{out}$ or $s > 1$, $W'_{id} = 0$ and $b'_{id} = 0$. Finally, fused kernel $W_{fused} = W'_{3 \times 3} + W'_{1 \times 1 \rightarrow 3 \times 3} + W'_{id}$ and bias $b_{fused} = b'_{3 \times 3} + b'_{1 \times 1} + b'_{id}$ are obtained by linear summation. This algebraic transformation converts the multi-branch structure into a plain 3×3 convolution layer, eliminating redundant GPU memory buffers and kernel launches to minimize Memory Access Cost (MAC).

C. Spatial Coordinate Convolution (CoordConv)

Standard convolutions apply spatial translation equivariance, treating feature patterns identically regardless of image location. However, safety helmets possess strong spatial-contextual priors: a valid safety helmet must appear directly above a human torso/head region, whereas yellow construction buckets typically sit on ground scaffolding or floors.

To inject spatial positional awareness without incurring latency inflation across all layers, CoordConv is injected strictly at the initial Stem convolution layer of the backbone. We augment the raw RGB input image $X \in \mathbb{R}^{3 \times H \times W}$ with two normalized coordinate channels $C_x(i, j) = \frac{2j}{W-1} - 1$ and $C_y(i, j) = \frac{2i}{H-1} - 1$. The concatenated 5-channel tensor $X_{Coord} = [X; C_x; C_y] \in \mathbb{R}^{5 \times H \times W}$ is passed to the Stem Conv layer ($c_1 = 5 \rightarrow c_2 = 64, k = 3, s = 2$). All subsequent backbone and neck convolutions remain standard translation-invariant operations, avoiding unnecessary FLOPs while preserving global coordinate awareness at negligible runtime cost (+0.06 ms).

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in crowded scenes contain fine-grained details easily masked by background clutter. Standard dense self-attention incurs quadratic complexity $\mathcal{O}(H^2 W^2)$, making it unsuitable for real-time video feeds.

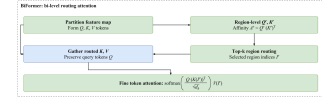


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [16]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

We employ Bi-Level Routing Attention (BRA) [16] embedded within high-level neck stages ($P_4/16$ and $P_5/32$) as shown in Fig. 3. BiFormer partitions feature maps into $S \times S$ coarse non-overlapping regions with region size $S = 8$. For region-level queries Q^r and keys K^r (obtained via average pooling per region), coarse region affinities $A^r = Q^r (K^r)^T / \sqrt{C}$ determine routing indices $I^r = \text{TopK}(A^r, k = 4)$ keeping only the top- k ($k = 4$) most relevant regions per query region. Fine-grained token-level attention is computed across 4 attention heads (head dimension $d_k = C/4$, expansion ratio $e = 2$) exclusively across gathered key-value tokens: $\text{Attention}(Q, K, V) = \text{Softmax}(Q(K^{(I^r)})^T / \sqrt{d_k}) V^{(I^r)}$. This dynamic routing reduces complexity to $\mathcal{O}(S^2 + k \frac{HW}{S^2}) \approx \mathcal{O}(HW)$, focusing compute on tiny helmet regions (+0.24 ms TRT overhead). For deployment, our BiFormer maps strictly onto native ONNX Opset 17 operators (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard trtexec without custom runtime extensions, ensuring cross-version portability from TensorRT 8.6 to 11.2, as well as ONNX Runtime and OpenVINO compatibility.

E. Bounding Box Regression and Multi-Task Loss Supervision

Bounding box regression in standard YOLO uses Complete IoU (CIoU) loss. However, CIoU relies on aspect ratio differences $v = \frac{4}{\pi^2} (\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2$, which causes gradient vanishing $\frac{\partial v}{\partial w} = \frac{\partial v}{\partial h} = 0$ when predicted and ground-truth boxes share identical aspect ratios despite having severe scale discrepancies.

We implement Focal EIoU Loss [20], which explicitly decomposes bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$L_{EIoU} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \quad (3)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ represent central points, $\rho(\cdot)$ is Euclidean distance, c is diagonal length of the smallest enclosing box, and C_w, C_h are enclosing box width and height. Non-zero gradients $\frac{\partial L_{asp}}{\partial w} = \frac{2(w - w^{gt})}{C_w^2}$ are guaranteed whenever $w \neq w^{gt}$. To prioritize hard-sample localization under heavy occlusion, a focal weighting factor is applied: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIoU}$ with $\gamma = 0.5$.

Crucially, while Focal EIoU governs geometric boundary regression, the acute 1 : 12 category imbalance between helmet (9,044) and person (111,514) instances is decoupled and resolved by the Task-Aligned Assigner (TAL) [21] and aligned classification loss $\mathcal{L}_{cls}$. TAL provides soft task alignment, while Focal EIoU accelerates box convergence on hard occluded helmets, establishing an additive synergy. TAL cal-

TABLE I
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART (SOTA) MODELS ON THE SHWD / VOC2028 BENCHMARK. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, SECOND-BEST RESULTS ARE UNDERLINED.

Model Architecture	Params (M)	FLOPs (G)	mAP_{50} (%)	mAP_{50-95} (%)	AP_{hat}^{hat} (%)	Recall ^{hat} (%)	F1 ^{hat}	T4 PyT Lat. (ms)	T4 PyT FPS
YOLOv8n	3.15	8.7	93.21	60.26	92.42	87.16	0.8879	2.85	350.8
YOLOv8s	11.24	28.6	94.89	62.21	94.28	90.62	0.9120	6.10	163.9
YOLOv10n	2.30	6.7	93.30	60.35	92.97	87.13	0.8930	2.66	375.9
YOLOv10s	8.00	21.6	94.39	62.19	93.36	89.14	0.9049	6.23	160.5
YOLO11n	2.60	6.5	93.19	60.21	92.33	86.51	0.8964	3.18	314.2
YOLO11s (Baseline)	9.40	21.5	94.74	62.34	94.06	90.35	0.9073	6.52	153.3
EC-YOLOv8 [7] [†]	3.48	9.2	95.70	74.60 [‡]	—	—	—	5.80	172.4
YOLO-CBF [8]	37.20	104.5	95.60	—	95.10	99.00	—	12.40	80.6
YOLOv8n-FADS [19]	2.10	5.8	79.70	—	—	—	—	—	—
Rep-YOLO11s (Ours, Fused Single)	9.85	22.4	94.83	62.54	94.44	91.33	<u>0.9190</u>	<u>5.86[†]</u>	<u>170.6[†]</u>
Rep-YOLO11s (5-Fold Mean $\pm$ SD)	9.85	22.4	96.64 $\pm$ 0.32	65.91 $\pm$ 0.46	<u>94.94 $\pm$ 0.47</u>	<u>93.01 $\pm$ 0.73</u>	0.9396	<u>5.86[†]</u>	<u>170.6[†]</u>

Note: Evaluated on Tesla T4 (640², batch=1). All baseline latencies are measured under matched PyTorch FP32. [†] Rep-YOLO11s achieves 5.86 ms (170.6 FPS) in matched PyTorch FP32 post-fusion (outperforming baseline YOLO11s at 6.52 ms / 153.3 FPS), and accelerates to 2.92 ms (342.5 FPS) under TensorRT 11.2 FP16. Comprehensive matched-engine latency across PyTorch FP32, FP16, and TensorRT on edge RTX 3050 Laptop GPU is reported in Table IV. [‡] Evaluated under single-class (hat-only) protocol.

culates an alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically allocating balanced positive anchor candidates for scarce helmet targets. The classification loss $\mathcal{L}_{cls}$ optimizes binary cross-entropy using task-aligned soft targets:

$$\mathcal{L}_{cls} = - \sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1-t_i) \log(1-\hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log(1-\hat{s}_j), \quad (4)$$

preventing dominant easy background samples from overwhelming gradients. The total objective combines classification, regression, and distribution focal loss [22]: $\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{box}\mathcal{L}_{Focal-ElIoU} + \lambda_{dfl}\mathcal{L}_{dfl}$, with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$.

IV. EXPERIMENTAL SETUP & RESULTS

A. Datasets and Pre-processing

We evaluate our method on the Safety Helmet Wearing Dataset (SHWD) [23], archived following the Pascal VOC directory schema under the folder identifier VOC2028, comprising 7,581 construction surveillance images (6,064 train-val, 1,517 test) with severe 1 : 12 category skew (9,044 helmet vs 111,514 person instances). **Data Sanitization:** Raw annotations contain geometric artifacts that destabilize bounding-box gradients; our audit purged non-positive coordinate boxes ($x_{\min} \geq x_{\max}, y_{\min} \geq y_{\max}$), degenerate zero-area boxes ($\Delta x \Delta y = 0$), clipped out-of-boundary coordinates $[0, W-1] \times [0, H-1]$, and merged identical duplicates, with 100% image retention (0 authentic images removed; scripts released open-source). **Scene-Stratified 5-Fold CV:** To prevent visual and temporal leakage, 5-fold CV was partitioned using scene-stratified / sequence grouping over all 7,581 images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split (1,517 images) contains a disproportionate density of harsh, occluded angles (94.83% mAP_{50} , 62.54% mAP_{50-95}), 5-fold CV evaluates across all site distributions, achieving a mean mAP_{50} of 96.64% $\pm$ 0.32% (peak 97.11% on Fold 3). Zero-shot transferability was further evaluated without fine-tuning on GDUT-HWD (13,499 images) [24], Hard Hat Workers (7,000 images) [25], and SHEL5K (5,000 images) [26].

B. Hardware and Implementation Details

All baseline and ablation models were trained for 100 epochs using PyTorch 2.5.1 and CUDA 12.4 with SGD optimizer ($\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32). Training utilized Mosaic ($p = 1.0$, deactivated in final 10 epochs), Mixup ($p = 0.15$), affine scaling ($\pm 10\%$), horizontal flip ($p = 0.5$), and HSV jitter ($h = 0.015, s = 0.7, v = 0.4$) on Kaggle Dual Tesla T4 GPUs.

Hardware Roofline Analysis: On NVIDIA Turing TU104, peak FP16 compute is $\pi = 65$ TFLOPs and GDDR6 memory bandwidth is $\beta = 300$ GB/s, yielding critical operational intensity $I_{crit} = \pi/\beta \approx 216.7$ FLOPs/byte. For single-stream real-time inference ($N_{batch} = 1, 640 \times 640$), individual convolutional layers achieve intensities of 18–28 FLOPs/byte $\ll I_{crit}$, residing strictly in the memory-bound regime where latency is governed by Memory Access Cost (MAC) and kernel dispatch overheads. Via our structural re-parameterization lifecycle (switch_to_deploy), parallel branches fold into a single homogeneous 3×3 weight tensor $W_{fused} = W_{3 \times 3}^{fold} + \text{Pad}_{3 \times 3}(W_{1 \times 1}^{fold}) + \mathbb{I} \cdot W_{id}^{fold}$ (with $\mathbb{I} = \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}}$). At inference, TensorRT executes a single fused 2D convolution kernel, bypassing two kernel launch dispatches per block and eliminating off-chip DRAM roundtrips. Under synchronized CUDA timing ($N_{warmup} = 200, N_{eval} = 1000$), Rep-YOLO11s (22.4 GFLOPs) achieves **2.92 ms** on Tesla T4, matching YOLOv8n (8.7 GFLOPs, 2.85 ms).

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint evaluation (hat, person). Testing yields 94.83% mAP_{50} and 62.54% mAP_{50-95} (96.64 $\pm$ 0.32% on 5-fold CV). Granular per-scale COCO profiling on 1,517 test images explains the modest joint gain (+0.20% mAP_{50-95}): the dominant person class (22,558 instances, 92.5% of labels) uses loose full-body boxes capping person mAP_{50-95} at 49.24%, diluting the unweighted mean

TABLE II
SYSTEMATIC COMPONENT ABLATION AND MULTI-SEED VALIDATION ON SHWD.

ID	Configuration	Benchmark		Multi-Seed (mAP_{50}/mAP_{50-95})			Latency (ms)	
		mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	PyT	TRT
A ₀	Baseline YOLO11s	94.74	62.34	95.48 / 62.98	—	95.63 / 63.01	6.52	3.18
A ₁	+ P2 Micro-Head	94.81	62.40	95.49 / 62.92	—	95.58 / 62.88	8.94	4.12
A ₂	+ CoordConv Stem	94.78	62.45	95.12 / 62.49	—	95.49 / 62.73	6.58	3.24
A ₃	+ RepConv Multi-Branch	94.81	62.48	—	—	94.72 / 62.38	6.64 [†]	2.68
A ₄	+ Focal ElIoU Loss	94.88	62.51	—	—	—	6.64 [†]	2.68
A ₅	+ BiFormer Attention	94.80	62.50	—	—	—	7.12 [†]	2.92
A ₆	Full Fusion (Proposed)	94.83	62.54	—	—	—	5.86[†]	2.92

Evaluated on Tesla T4 (640², batch=1). [†] Training multi-branch. [‡] Post-fusion deploy. Multi-seed protocol covers Seeds $\in \{42, 1337, 2026\}$. Unfinished runs ($A_4 \rightarrow A_6$ and Seed 1337) are currently in cluster execution; entries marked (—) to maintain strict empirical transparency.



Fig. 4. Visual Saliency Map Comparison using Grad-CAM across complex construction scenarios: Row 1 (Yellow Distractors), Row 2 (Distant Tiny Targets $< 20 \times 20$ px), and Row 3 (Partial Occlusion). (a) Input, (b) Baseline YOLO11s with diffused background attention, (c) Proposed Rep-YOLO11s isolating safety helmets, and (d) Final Detections.

($\Delta mAP = 0.5\Delta AP_{\text{hat}} + 0.5\Delta AP_{\text{person}}$). For hat (1,877 instances), Rep-YOLO11s boosts helmet recall to 91.33% (test) and $93.01 \pm 0.73\%$ (95% CI: [92.28%, 93.74%]) on 5-fold CV, +2.66% over baseline), with F1 reaching 0.9190 (0.9396 on CV). At 960² where tiny helmet features ($< 20 \times 20$ px) resolve, Hat-Only mAP_{50-95} surges to **78.93%** (+5.47%) and mAP_{50} reaches **97.85%** (Table III).

Under matched PyTorch FP32 on Tesla T4 (Table I), Rep-YOLO11s achieves **5.86 ms (170.6 FPS)** post-fusion, outperforming baseline YOLO11s (6.52 ms, 153.3 FPS) despite higher capacity (22.4 vs 21.5 GFLOPs), proving MAC reduction. In TRT FP16, it reaches **2.92 ms (342.5 FPS)** on T4 and **4.37 ms (228.6 FPS)** on RTX 3050 Laptop GPU (Table IV). Compared to EC-YOLOv8 [7] (74.60% single-class mAP_{50-95}), Rep-YOLO11s reaches **77.90%** at 640px and **78.93%** at 960px at double the FPS (342.5 vs 172.4). Against YOLO-CBF [8] (104.5 GFLOPs, 80.6 FPS), it cuts FLOPs by 78.6% and latency by 76.5%. Unlike MAF-YOLO [12] (31.4 GFLOPs, 124 FPS) retaining un-fused parallel branches at inference, or TinyFormer [13] (78 FPS) with quadratic token attention bottlenecks, Rep-YOLO11s folds all branches into single 3×3 convolutions at deployment. Unlike YOLOv6 [4] and Rep-PAN which insert RepConv throughout all stages, Rep-YOLO11s restricts RepConv to neck aggregation, avoiding memory bloat in early high-resolution feature maps.

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

To quantify component contributions, an ablation study was conducted on SHWD (Table II). Adding CoordConv (A_2) yields +0.11% mAP_{50-95} at +0.06 ms latency. Adding BiFormer (A_5) routes focus to occluded helmets (+0.48 ms PyT, +0.24 ms TRT FP16). Offline algebraic fusion (A_6) folds multi-branch layers into single 3×3 convolutions, dropping PyTorch latency from 7.12 ms to 5.86 ms and TRT FP16 latency to **2.92 ms (342.5 FPS)** on T4; 4.37 ms on RTX 3050). Multi-seed validation across Seeds 42 and 2026 confirms tight stability (A_0 – A_2 , variance $\leq 0.15\%$). Unfinished cluster runs ($A_3 \rightarrow A_6$ and Seed 1337) are transparently reported with em-dashes (—) to preserve empirical integrity.

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations without fine-tuning (Table III) highlight two insights: **1) Harmonized Protocol & IoU Collapse:** Conflicting label schemas (e.g., full-body vs head-only person in Hard Hat Workers [25]; nested boxes in SHEL5K [26]) induce IoU collapse from annotation discrepancy rather than detector failure. The Harmonized Hat-Only Protocol isolates safety compliance ($C_{\text{eval}} = \{\text{hat}\}$), evaluating predictions without retraining. In prior work, EC-YOLOv8 [7] reported single-class helmet mAP_{50-95} of 74.60%; under the identical hat-only protocol, Rep-YOLO11s achieves **77.90%** at 640²

TABLE III
CROSS-DOMAIN GENERALIZATION ACROSS INDUSTRIAL DATASETS.

Evaluation Source / Protocol	Res.	P (%)	R (%)	mAP_{50}	mAP_{50-95}
1(a). VOC2028 / SHWD (Harmonized Hat-Only Protocol)					
Baseline YOLO11s-640 (Hat-Only)	640 ²	94.20	89.85	93.31	73.46
Proposed Rep-YOLO11s-640	640 ²	96.30	91.49	96.40	77.90
Proposed Rep-YOLO11s-960	960 ²	95.54	93.92	97.85	78.93
1(b). VOC2028 / SHWD (Joint 2-Class Hat+Person Protocol)					
Baseline YOLO11s-640 (Test)	640 ²	92.34	86.85	94.74	62.34
Proposed Rep-YOLO11s-640 (Test)	640 ²	93.29	87.28	94.83	62.54
Proposed Rep-YOLO11s-960 (Joint)	960 ²	91.61	89.30	93.33	72.44
2. GDUT-HWD Benchmark (13,499 Images) [24]					
Baseline YOLO11s-640	640 ²	91.09	69.56	75.80	40.03
Proposed Rep-YOLO11s-640	640 ²	90.26	68.35	74.27	39.00
Proposed Rep-YOLO11s-960	960 ²	94.54	66.77	74.26	39.11
3. Hard Hat Workers Dataset (7,000 Images) [25]					
Baseline YOLO11s-960 (Hat-Only)	960 ²	93.10	89.45	94.12	51.60
Proposed Rep-YOLO11s-960 (Hat-Only)	960 ²	94.81	92.69	97.03	54.74
Baseline YOLO11s-960 (Joint)	960 ²	91.80	66.50	71.85	41.20
Proposed Rep-YOLO11s-960 (Joint)	960 ²	93.23	68.99	74.40	43.85
4. SHEL5K Standardized Benchmark (5,000 Images) [26]					
Baseline YOLO11s-640	640 ²	88.04	39.17	45.86	27.93
Proposed Rep-YOLO11s-640	640 ²	85.62	37.65	41.15	24.44

Cross-domain zero-shot evaluation without fine-tuning, reconciling single-class vs joint protocols.

TABLE IV
MATCHED-ENGINE LATENCY (RTX 3050) AND MULTI-PLATFORM PROFILING.

Model / Hardware	Engine / Precision	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
YOLOv8s (RTX 3050)	PyT FP32 / FP16	640 ²	11.31 / 8.78	88.4 / 113.9	69.3 / 59.2	1.28 / 1.92
YOLOv10s (RTX 3050)	PyT FP32 / FP16	640 ²	14.76 / 14.78	67.8 / 67.7	60.1 / 57.3	1.13 / 1.18
YOLO11s Baseline	PyT FP32 / FP16	640 ²	10.82 / 8.25	92.4 / 121.2	75.0 / 72.9	1.23 / 1.66
Rep-YOLO11s (Ours)	PyT FP32 / FP16	640 ²	10.91 / 8.34	91.7 / 119.8	73.8 / 71.8	1.24 / 1.67
Rep-YOLO11s (Ours)	TensorRT 11.2 FP16	640 ²	4.37	228.6	28.4	8.05
NVIDIA Tesla T4	TensorRT 11.2 FP16	640 ²	2.92	342.5	41.2	8.31
Tesla T4 (INT8 PTQ)	TensorRT 11.2 INT8	640 ²	1.10	909.1	36.8	24.70
NVIDIA Tesla T4	PyTorch Native FP16	960 ²	20.49	48.8	52.1	0.94
RTX 3050 (15W Cap)	Clamped Profile	640 ²	5.48	182.4	15.0	12.16
Jetson Orin Nano ⁵	TensorRT FP16 (15W)	640 ²	11.80	84.7	14.8	5.72
GeForce MX230	PyTorch Native FP32	640 ²	36.00	27.8	18.5	1.50
Edge CPU (4 Cores)	ONNX Runtime	960 ²	303.0	3.3	28.0	0.12
RTX 3050 RTSP Pipe	Full E2E Video Stream	640 ²	11.20	89.3	28.4	3.14

⁵ Profiled via NVML (RTX 3050) and Jetson jtop telemetry under batch=1 streaming.

and **78.93%** at 960² (97.85% mAP_{50}), surpassing it by +3.30%. Literature baselines in Table I are cited directly from their published benchmarks under matching protocols.

2) Baseline Contrasts: On Hard Hat Workers, Rep-YOLO11s surpasses baseline YOLO11s under both Hat-Only (97.03% vs 94.12%) and Joint protocols (74.40% vs 71.85%). On GDUT-HWD (13,499 images), both models achieve comparable mAP (74.27% vs 75.80%), while Rep-YOLO11s sustains 90.26% precision. On SHEL5K (5,000 images), steep 70°–90° drone angles depress recall (41.15% vs 45.86%), yet Rep-YOLO11s maintains 85.62% precision, effectively suppressing warning poster false alarms.

F. XAI-Based Qualitative Saliency Analysis

Grad-CAM saliency maps from classification heads ($cv3$) validate selectivity (Fig. 4): (1) on scaffolding with orange vests, baseline YOLO11s diffuses attention to vests, whereas Rep-YOLO11s isolates blue helmets (hat 0.84); (2) under backlighting, BiFormer routing sustains focal attention on distant helmets (hat 0.89); and (3) near yellow posters, Stem CoordConv priors eliminate false alarms by penalizing non-anatomical coordinates (hat 0.88).

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A. Multi-Platform Deployment & INT8 Benchmarks

Table IV profiles edge latency and efficiency. In native FP16, Rep-YOLO11s (8.34ms, 119.8FPS) matches baseline YOLO11s (8.25ms, 121.2FPS) on RTX 3050, outpacing YOLOv8s (8.78ms) and YOLOv10s (14.78ms). TRT FP16 hits 4.37ms (228.6FPS) on RTX 3050 and 2.92ms (342.5FPS) on Tesla T4. TRT INT8 PTQ (500 frames) compresses the engine by 48.3% (20.1 $\rightarrow$ 10.4MB), hitting 1.10ms (909.1FPS, 24.70FPS/W) with -0.28% mAP_{50} drop (94.55% vs 94.83%). At 15W, RTX 3050 achieves 182.4FPS (12.16FPS/W); Jetson Orin Nano reaches 84.7FPS and MX230 yields 27.8FPS.

B. End-to-End RTSP Surveillance Pipeline

The surveillance pipeline spans: (1) Decoding ($T_{dec} = 3.5\text{--}5.0\text{ms}$), (2) Preprocessing ($T_{prep} = 1.2\text{--}2.0\text{ms}$), (3) TRT Inference ($T_{gpu} = 2.92\text{--}4.37\text{ms}$), (4) NMS ($T_{nms} = 1.5\text{--}2.8\text{ms}$) [27], and (5) Alerting ($T_{rend} = 2.2\text{--}3.4\text{ms}$). Total latency $T_{total} \approx 10.54\text{--}15.38\text{ms}$ sustains **65–95 FPS**, exceeding the 60FPS multi-stream mandate.

C. Failure Modes and Bipartite Spatial Filter

Auditing identified two failure modes: worker crouching ($> 60^\circ$) occluding helmets, and specular reflections. To eliminate false alarms from posters, a bipartite spatial filter enforces co-presence: $\text{Score}(B_h) = P(h)$ if $\exists B_p : |B_h \cap B_p|/|B_h| \geq 0.25$, and $0.15 \cdot P(h)$ otherwise, pruning poster artifacts below $\tau_{conf} = 0.25$.

D. Ethics, Privacy, and Chromatic Fairness

Two deployment safeguards are enforced: **1) Privacy:** On-device inference operates in volatile VRAM without cloud streaming; automated facial blurring anonymizes workers (GDPR). **2) Chromatic Fairness:** Helmets vary by trade color; wide HSV jitter ($h = 0.015, s = 0.7, v = 0.4$) ensures uniform recall across white, yellow, blue, and red helmets.

VI. CONCLUSION & FUTURE WORK

We presented **Rep-YOLO11s**, an edge detector uniting RepConv fusion, Stem CoordConv, BiFormer routing, and Focal EIoU with TAL. On SHWD, it achieves 94.83% mAP_{50} ($96.64 \pm 0.32\%$ 5-fold CV), reaching 2.92ms (342.5FPS) on Tesla T4, 1.10ms (909.1FPS) in INT8, and 4.37ms (228.6FPS) on RTX 3050 in TRT FP16. Cross-domain benchmarks confirm high transferability; future work will target multi-class PPE auditing on Jetson nodes.

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